

# SYNOPSIS

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**Project Title:** AN IMAGE PROCESSING AND HYBRID DEEP LEARNING SYSTEM FOR DIABETIC RETINOPATHY SEVERITY DETECTION

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## Abstract:

Diabetic retinopathy (DR) is a major cause of global blindness, affecting millions worldwide and requiring accurate severity grading for timely treatment. This study proposes a hybrid deep learning framework to improve DR classification by integrating advanced feature extraction, optimization, and classification techniques. The system utilizes the Kaggle EyePACS dataset containing 35,126 retinal images categorized into five severity levels: normal, mild, moderate, severe, and proliferative DR. To address class imbalance and enhance data diversity, data augmentation techniques such as rotation, flipping, and cropping are applied, significantly increasing the dataset size. Deep features are extracted using a fusion of GoogleNet and ResNet architectures, capturing rich and complementary representations. These features are further optimized using Adaptive Particle Swarm Optimization (APSO) to improve discriminative capability. Finally, a Radial Support Vector Machine (RSVM) is employed for accurate classification of disease severity. The proposed approach demonstrates strong potential for reliable clinical decision support and early diagnosis, contributing to the prevention of vision loss.

## Specific Contribution:

- Worked on diabetic retinopathy classification using deep learning and machine learning techniques.
- Trained pretrained GoogLeNet and ResNet18 models using transfer learning and fine-tuning on the EyePACS retinal image dataset.
- Used the trained models as deep feature extractors to generate meaningful retinal image feature representations.
- Applied Adaptive Particle Swarm Optimization (APSO) to refine the extracted features and remove redundant information.
- Classified the optimized features using a Radial Support Vector Machine (RSVM) to improve diabetic retinopathy severity prediction performance.

### **Specific Learning:**

- Learned about diabetic retinopathy (DR) and understood the importance of early severity classification in preventing vision loss and supporting timely treatment.
- Gained experience in training pretrained deep learning models such as GoogLeNet and ResNet18 for medical image classification, along with deep feature extraction, feature fusion, APSO-based feature optimization, and RSVM-based diabetic retinopathy severity classification.
- Learned how deep learning and machine learning techniques can be applied in healthcare applications for automated disease detection and medical image analysis.

### **Technical Limitations & Challenges Faced:**

#### **1. APSO Optimization Complexity:**

Adaptive Particle Swarm Optimization required careful tuning of parameters such as particle size, iteration count, and feature selection strategy. Improper parameter settings sometimes led to slow convergence and inconsistent feature selection results.

#### **2. SVM Hyperparameter Sensitivity:**

The performance of the Radial SVM classifier was highly dependent on selecting suitable values for parameters such as C and gamma, requiring extensive hyperparameter tuning.

#### **3. High Computational Cost:**

Training deep learning models, extracting deep features, running APSO optimization, and performing GPU-based SVM classification required high computational power and memory usage.

#### **4. Risk of Overfitting:**

Because of complex feature representations and augmented training samples, the model occasionally showed signs of overfitting on the training dataset.

**Keywords:** Diabetic Retinopathy, Feature Extraction, GoogLeNet, ResNet18, Adaptive Particle Swarm Optimization (APSO), Radial Support Vector Machine (RSVM), Medical Image Analysis

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